

Model Evaluation

100 rows

Train 70%
80%
800

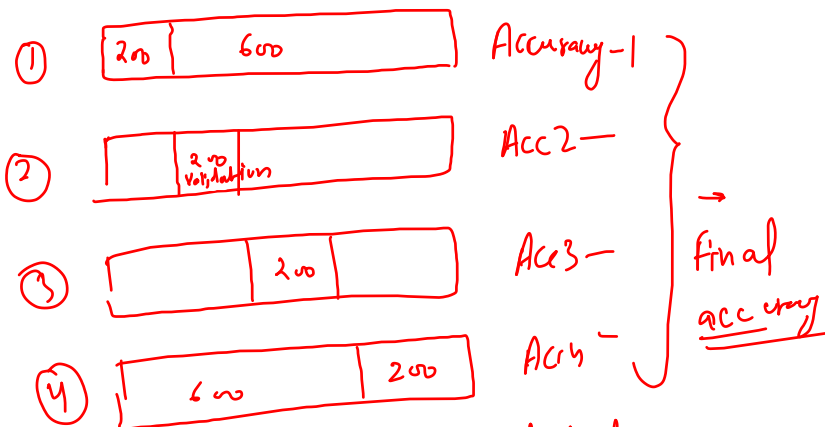
Validation
 ↓

Test 20-30%
200

Cross Validation

$$CV = 4$$

$$\frac{800}{4} = 200$$



Hyperparameter Tuning

max_depth = max_leafnode =

clf = DecisionTreeClassifier (Split =

2, 2, 'gini')

3, 1, 'm'

Grid Search CV

Random Search CV

Cross Validation

Machine Learning

Hyperparameter Tuning Techniques

CV =

max_depth

[2, 3, 4, 5]

split

[4, 1, Entropy]

max_leafnode

[2, 3]

500

2, gini, 2

4

x

2

x

2

= 16

2, Entropy, 2

4 x 2

A = 2, 3

→

2

x

3

⇒

(A, B)

B = 6, 8, 9

-

3

(2, 6)

(3, 6)

(2, 8)

(3, 8)

(2, 9)

(3, 9)

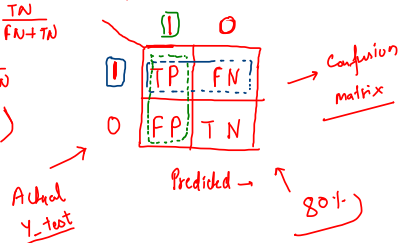
Classification Metrics

- ① Accuracy = $\frac{TP+TN}{TP+FN+FP+TN}$
- ② Precision = $\frac{TP}{TP+FP}$, $\frac{TN}{FN+TN}$
- ③ Recall = $\frac{TP}{TP+FN}$, $\frac{TN}{FN+TN}$
- ④ F-β Score (β = 0.5, 1, 2)
- ⑤ ROC - AUC
- ⑥ PRC - AUC

Confusion Matrix

Binary Classification

0 - True
1 - False

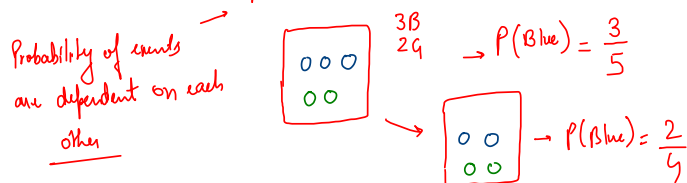


Multiclass Classification Evaluation

Naive Bayes Classifier

Probability

- Independent → $P(\text{Head}) = \frac{1}{2}$
- Dependent →



Independent Events - A and B

$$P(A \text{ and } B) = P(A) * P(B) \quad P(B/A)$$

$$\Rightarrow P(A) * P(B/A)$$

$$P(B \text{ and } A) = P(B) * P(A/B)$$

$$P(A \text{ and } B) = P(B \text{ and } A)$$

$$P(A) * P(B/A) = P(B) * P(A/B)$$

$$P(A) = \frac{P(B) * P(A/B)}{P(B/A)}$$

$$P(A/B) = \frac{P(A) * P(B/A)}{P(B)}$$

x_1	x_2	x_3	y

$$\Rightarrow P(y, x_1, x_2, x_3) = \frac{P(y) * P(x_1, x_2, x_3/y)}{P(x_1, x_2, x_3)}$$

Wind	Outlook	Weather	Play
Mild	Overcast	Sunny	Yes
Windy	Rainy	Rain	No
Mild	-	-	Yes

$$P(y, x_1, x_2, x_3) = \frac{P(y) * P(x_1/y) * P(x_2/y) * P(x_3/y)}{P(x_1) * P(x_2) * P(x_3)}$$

Windy, Overcast, Rainy, Mild, Yes, No

Yes

No

$P(Y) = 0.08$

$P(N) = 0.35$

